

EX ANTE ANALYSIS · FULL WALKTHROUGH · 2026-09 MODEL BASIS

Mapping the public and private returns to Enhanced Rock Weathering in Sub-Saharan Africa

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A spatially explicit, decomposed profitability model — 44 SSA countries · 23 crops · per-pixel grid

Working draft for Agricultural Economics — coauthor review. All numbers from the 2026-09-01 run:

Arrhenius weathering kinetics ($E_a = 68.8$ kJ/mol) · pH optimum 6.0 · net-export CDR accounting · ESROC-verified sources

THE PROPOSITION

ERW pays the farmer twice

Spreading finely ground basalt on cropland couples two value streams that are usually separate:

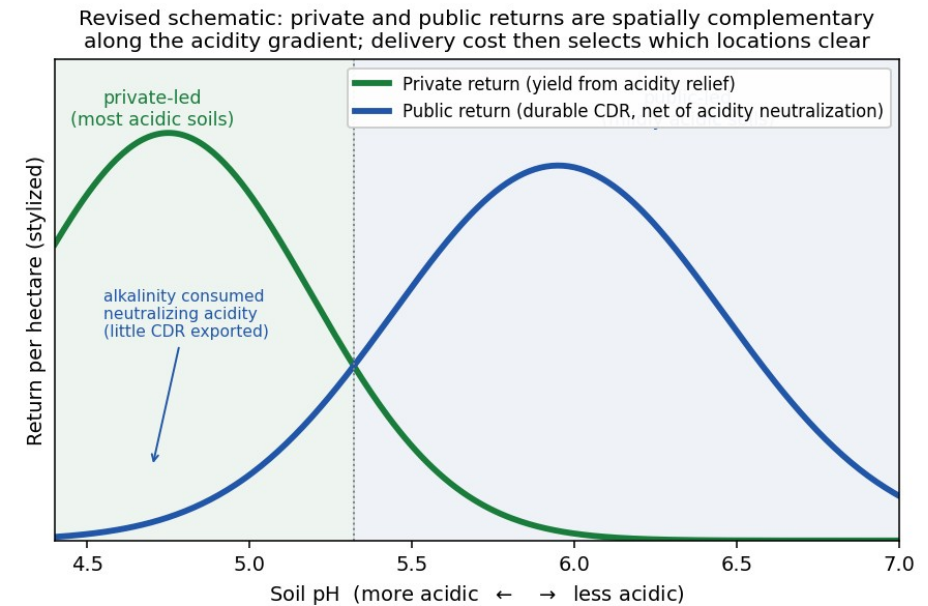
PRIVATE — agronomic

Alkalinity neutralizes soil acidity like agricultural lime, raising yields. Accrues to the farmer.

PUBLIC — carbon (CDR)

Weathering durably removes atmospheric CO₂, monetizable through voluntary carbon markets (VCM).

Central question: where does each return justify deployment — and where does neither suffice alone, yet together they do?



Realized returns are additionally gated by the delivery-cost gradient (distance to basalt): within each band, only low-cost, basalt-proximate pixels clear.

Six research questions

Q1

Where does the private (yield) return alone justify ERW investment?

Q2

Where does the public (carbon) return alone justify it?

Q3

Where do the two coincide — the intersection?

Q4

Where does neither stream alone suffice, but the combined margin is positive?

Q5

How robust is the deployable geography to parameter assumptions?

Q6

What would shift the public (carbon) envelope outward?

A decomposed per-pixel gross margin

$$\pi = R_{\text{agro}} + R_{\text{CDR}} - C$$

gross margin = private agronomic return + public carbon return - full delivered cost

Resolution

~0.083° grid × 44 SSA countries × 23 SPAM crops, area-weighted to each pixel.

Three streams

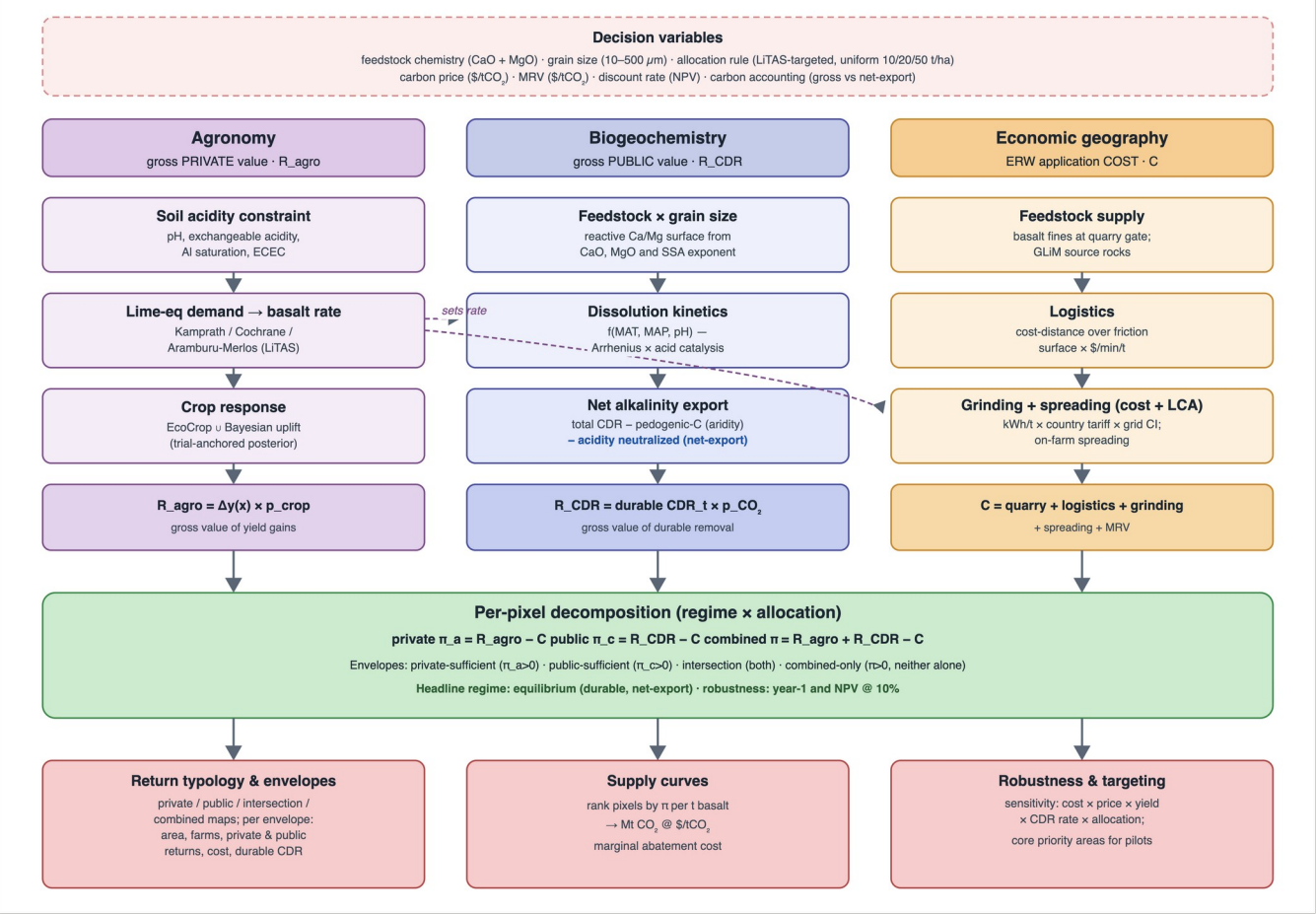
Agronomic yield uplift, durable CDR credit, and a fully decomposed delivered cost.

Decision object

Each pixel is classified by which stream(s) make the margin positive — not a single supply curve.

Regimes

Evaluated at year-1, NPV @ 10%, and equilibrium steady state (the headline).



INPUTS

Public data layers behind every pixel

SOIL

SoilGrids

pH, ECEC, texture on cropland

CROPS

SPAM v2

23 crop distributions & production

AGRONOMY

EcoCrop

crop pH-suitability response

FEEDSTOCK

GLiM

basalt source geology

LOGISTICS

MAP friction 2019

travel-time cost surface (Weiss et al. 2020)

CLIMATE

WorldClim

temperature & moisture for weathering

PRICES

FAOSTAT

2016–20 producer-price medians

LIME REQ.

LiTAS

lime-equivalent dose (targeting)

ACIDITY SINK

Merlos / Kamprath

standing & maintenance acidity for net-export

PART I

The assumption sets

Five modeling choices govern every result: delivered cost, carbon price, net-export CDR accounting, an over-liming private penalty, and kinetic weathering saturation.

Cost is dominated by transport — a real lever

Decomposed cost

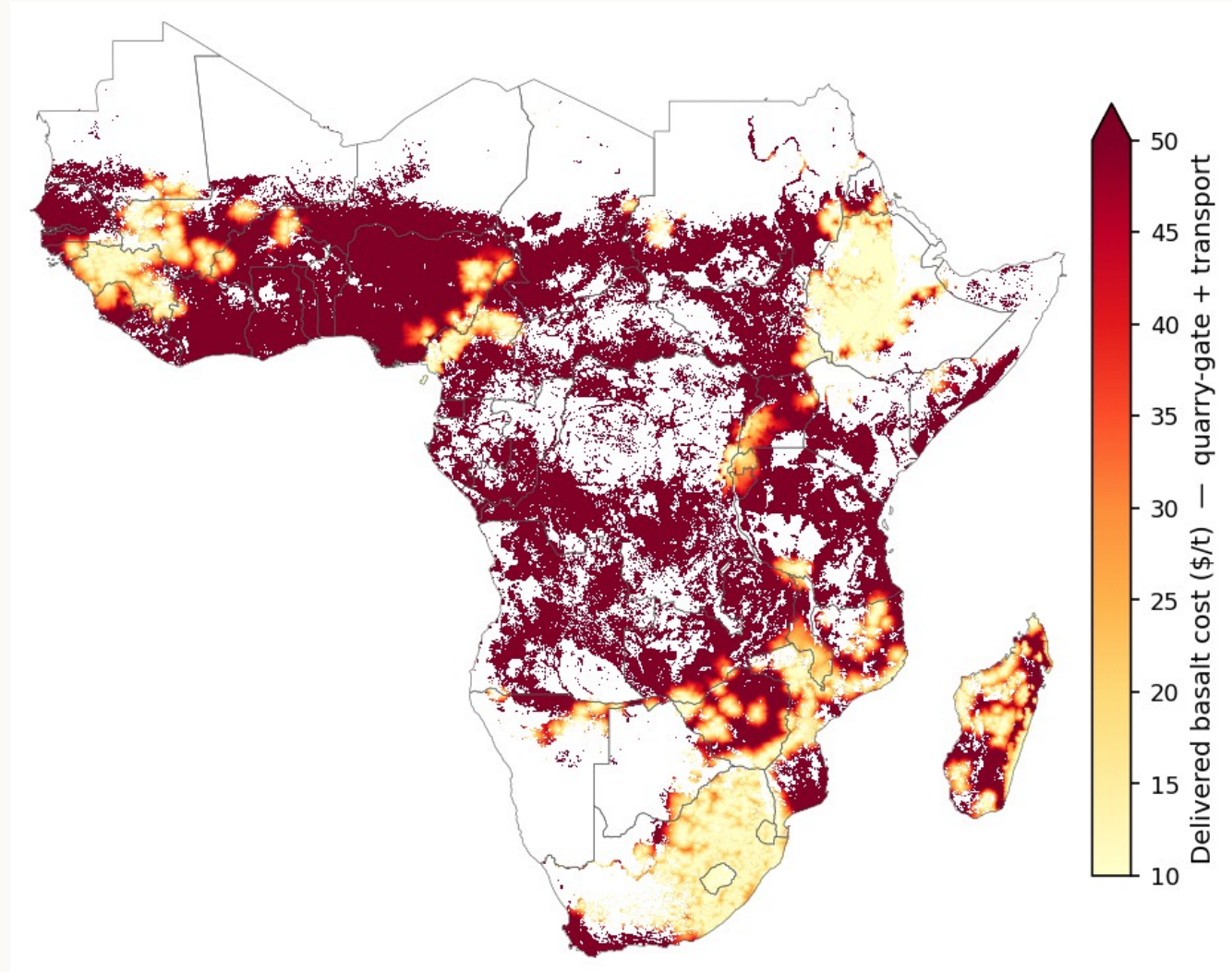
Quarry-gate + grinding + freight + spreading, built up per pixel from the friction surface; MRV netted from the carbon price.

Transport binds

Mean 55% (median 66%) of delivered cost. Remote acid soils are priced out even where the agronomy is strong.

The solar lever

Electrified / solar-assisted haulage cuts the variable transport term: -25% transport widens the private envelope 4.31 → 5.12 Mha, the public one only 1.95 → 2.07.



Delivered basalt cost, decomposed

Component	Cost per t basalt	Type	Basis / source
Quarry-gate feedstock	\$10 / t	fixed	basalt fines as aggregate by-product (cf. \$17.50/t US crushed-stone average, USGS)
Grinding (50 µm)	19.4 kWh/t × local tariff (≈\$1.9/t)	fixed*	Streffer et al. 2018 energy curve
Transport (haulage)	travel-time × \$0.04/min/t	variable	30 t truck over MAP friction surface; within SSA corridor tariffs (Teravaninthorn & Raballand 2009)
Spreading	\$8 / t	fixed	on-farm application
Delivered cost / t	quarry + grinding + transport + spreading	= sum	the delivered \$/t consumed per pixel
MRV (netted separately)	\$20 / tCO ₂	per tCO ₂	low end of the \$15–71/tCO ₂ supplier range (Mercer et al. 2024, LSE Grantham)

* grinding varies with the country electricity tariff; its life-cycle CO₂ (with transport and spreading) is separately deducted from gross removal.

The carbon return uses a net VCM price

$R_{CDR} = \text{durable CDR} \times (p - m)$

p = VCM carbon price (\$/tCO₂) · m = per-tonne MRV cost

Baseline is **\$150/tCO₂** — a conservative floor: 2024–25 durable-CDR transactions cluster at \$250–450/tCO₂ (CDR.fyi). Price, MRV and delivered cost move the public envelope only through the single ratio (p - m)/c.

The ladder is steeply convex: between \$250 and \$350 the deployable area jumps from 7 to 23 Mha.

\$/tCO ₂	Public Mha	% treated	Durable CDR (Mt)
\$100	1.40	4.3%	7.8
\$150	3.95	12.1%	20.4
\$250	7.00	21.5%	33.3
\$350	23.06	70.7%	100.6
\$500	27.98	85.8%	115.6

Treated base: 32.6 Mha under uniform-20. On the targeted dose the \$150 envelope is 1.95 Mha (11.1% of its 17.6 Mha treated base) — shares are not comparable across allocations.

Credit only carbon that is durably exported

$$\text{durable CDR} = \max(\text{gross} - F \cdot S, 0)$$

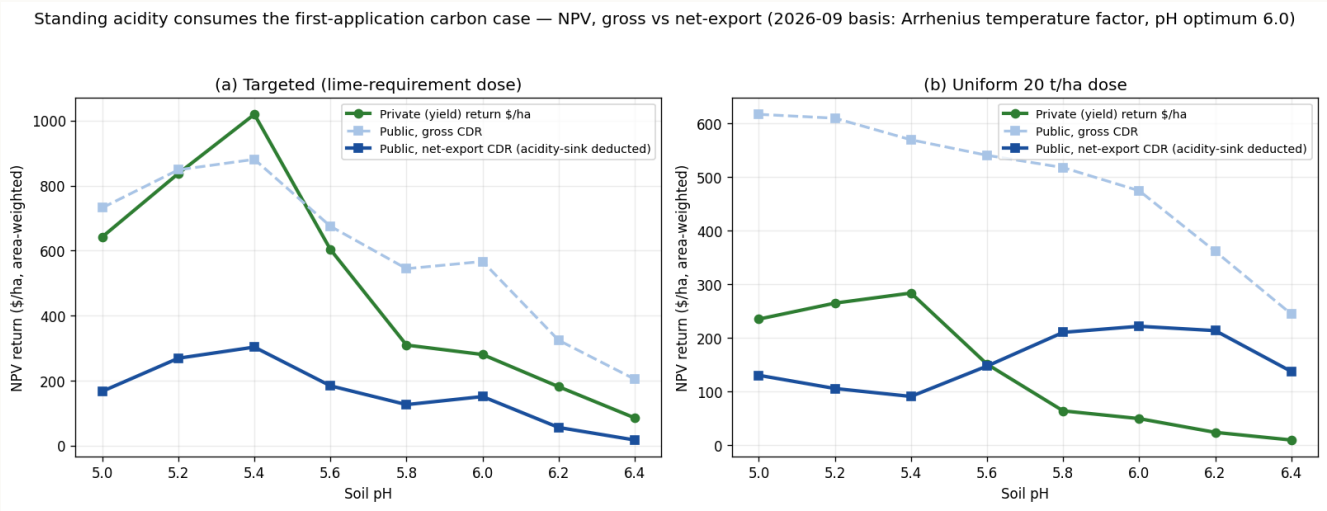
$F = 0.88 \text{ tCO}_2 \text{ per t CaCO}_3\text{-eq}$ · $S = \text{acidity sink (standing / maintenance)}$

Alkalinity spent neutralizing soil acidity re-releases its CO₂ — exactly as agricultural lime does — so it must be deducted (West & McBride 2005; Hamilton et al. 2007; Holden et al. 2024). A first-order sequential bound; exchange-stored cations can export later (Kanzaki et al. 2025).

≈70%

of credited removal is cut at first application once acidity neutralization is netted out

Durable removal is largely a **steady-state phenomenon** — equilibrium retention is 77% (targeted) to 90% (uniform-20). Hence the equilibrium headline.



Too much alkalinity can depress yield

Mechanism

Beyond the lime requirement, pH is pushed past the crop optimum; the EcoCrop response applies a yield haircut at the post-application pH.

Buffered shift

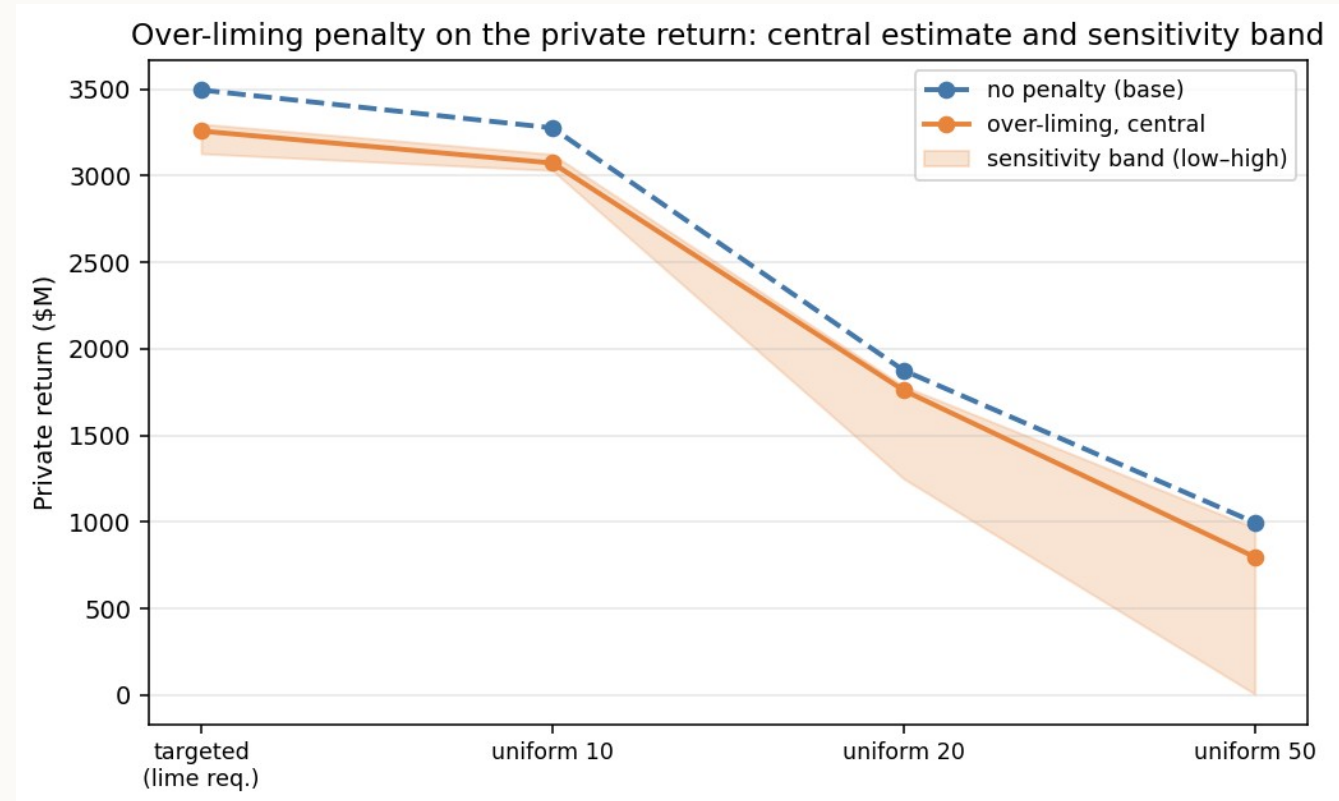
pH change is damped by soil buffering ($\beta = k\beta \cdot \text{ECEC}$); reacted alkalinity $A = \text{gross CDR} / F$.

Effect

Concentrated on high uniform doses: the central estimate trims the targeted private return ~7% (band 6–10%), but 20% at uniform-50 with a 3–99% band.

Status

The penalty parameters are stated assumptions — the ERW literature documents the risk only qualitatively (Swoboda et al. 2022).

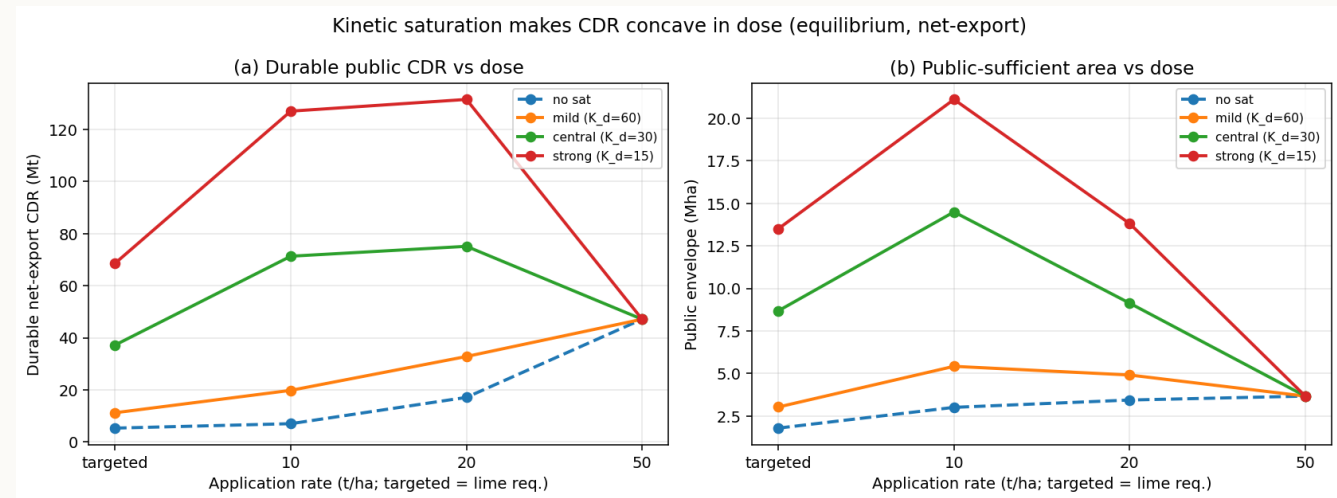


Weathering reacts less-than-linearly with dose

$$\text{sat}(D) = (K_d + 50) / (K_d + D)$$

effective reacted tonnage saturates in applied dose D, normalized at the Lewis (2021) 50 t/ha reference

Key finding. Saturation makes CDR concave in dose (giving an interior optimum), but it does not drive the public envelope to zero. Below the 50 t/ha reference, per-tonne reactivity is higher — so at targeted doses durable CDR actually rises. What binds the carbon case is $MAC > \text{price}$, not kinetics.



Temporal regimes × allocation rules

The headline is the **equilibrium steady state with net-export accounting** — the targeted dose leads the private and intersection geography; **uniform-20 leads the carbon case** (the lime-requirement dose is CDR-minimal by construction).

TEMPORAL REGIMES

Year-1

One-time dose & cost; durable CDR net of standing acidity. Most conservative on carbon.

NPV @ 10%

Discounted stream of relief and durable removal, phased over the 5-year residency.

Equilibrium

Steady-state durable removal against the maintenance dose — the headline regime.

ALLOCATION RULES

Targeted

LiTAS lime-equivalent dose — agronomically matched. Leads private & intersection.

Uniform 10 / 20 / 50 t/ha

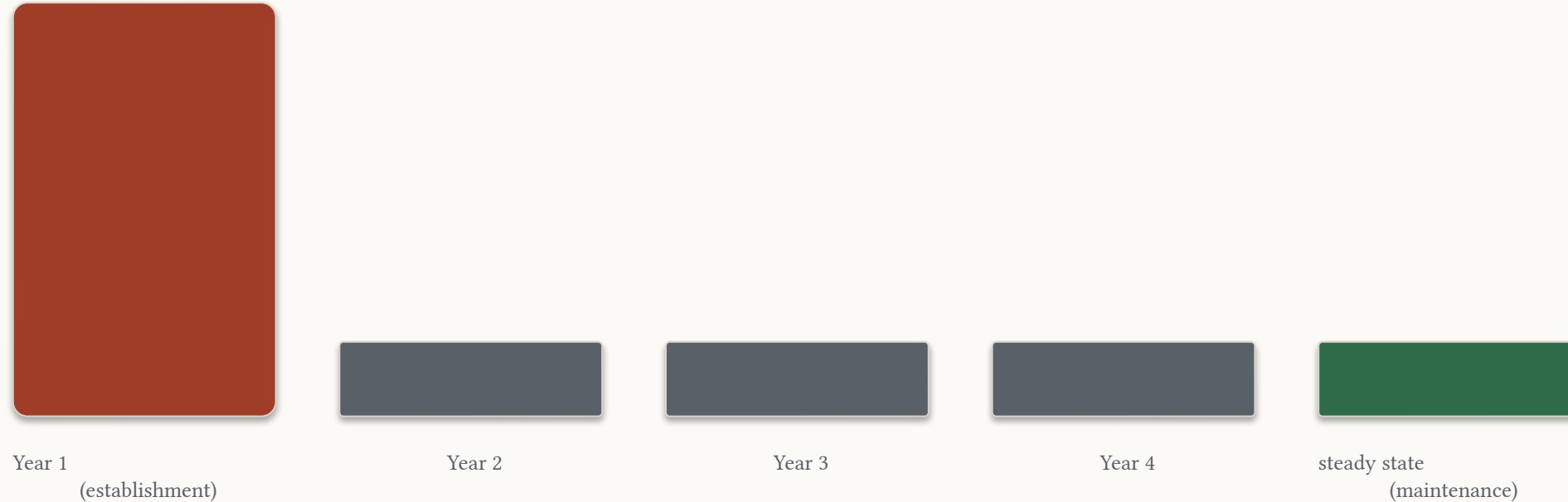
Fixed rates — trace how CDR and over-liming scale with dose. Uniform-20 leads the carbon case.

Weathering kinetics

Arrhenius temperature response ($E_a = 68.8$ kJ/mol) · unimodal pH factor peaking at 6.0 (2026-09 basis).

Establishment dose vs. steady-state maintenance

bar height $\propto$ basalt applied that period — charged at its full delivered cost; the yield benefit recurs every year once soils are held at target



EQUILIBRIUM — the headline

Full annual yield vs the fully-costed maintenance top-up, once soils are at target. Private envelope: $\approx$ \$3.5 B yield for $\approx$ \$1.74 B rock (~2:1 per cycle). Excludes the establishment dose.

YEAR-1 — establishment

One-time full lime-equivalent dose; standing acidity consumes most first-application alkalinity, so the carbon case is thin until steady state.

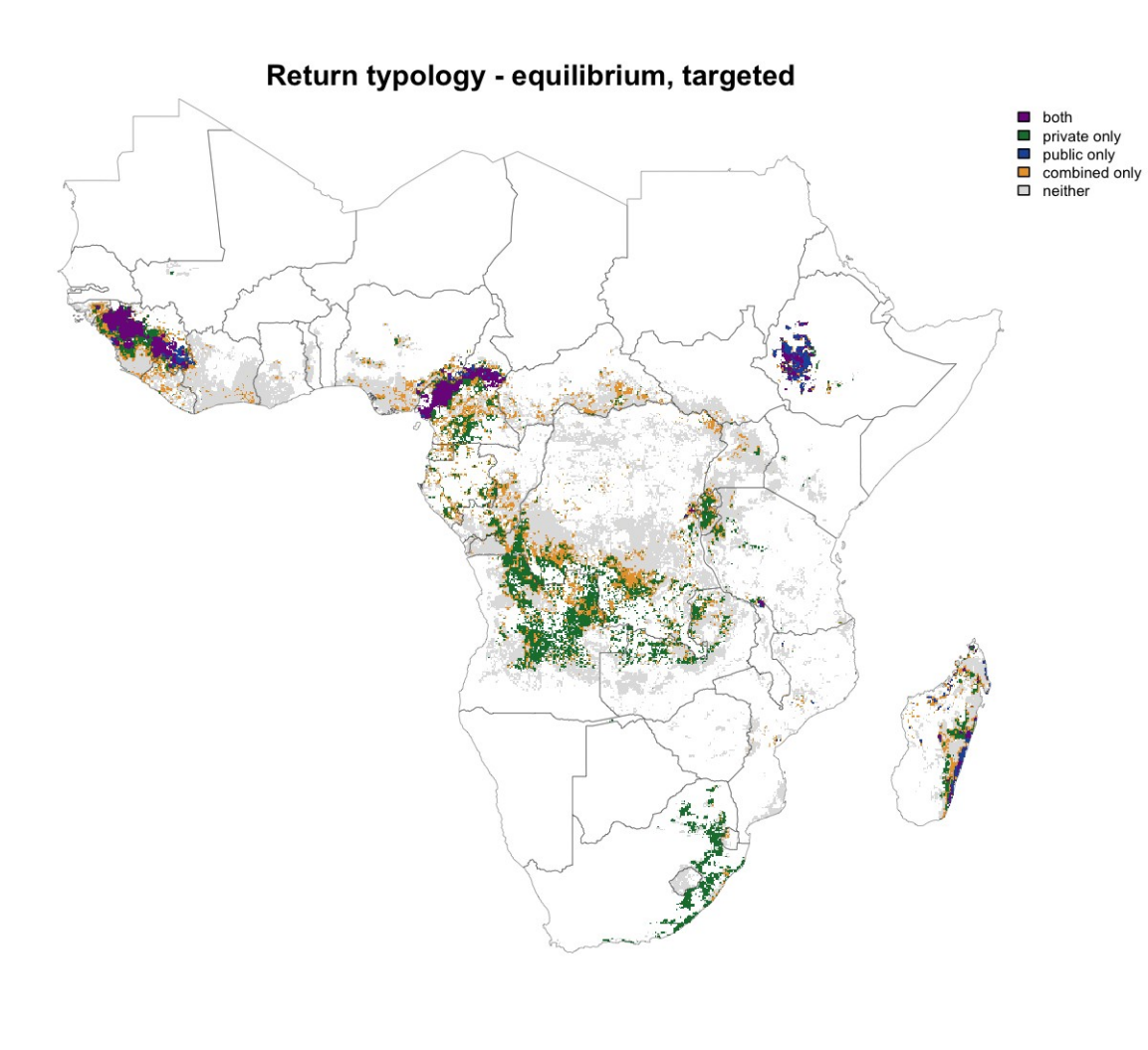
PART II

Results

Envelopes and the combined-only band · durable carbon and marginal abatement cost · what shifts the public frontier · optimal rates · the robust core geography.

A five-way pixel classification

Private-sufficient	$R_{agro} - C > 0$
Public-sufficient	$R_{CDR} - C > 0$
Intersection	both hold at the same pixel
Combined-only	neither alone, but $GM > 0$
Neither	$GM \leq 0$ — not deployable



Where yield alone pays

Land where the agronomic return covers the full delivered cost with no carbon credit at all.

4.3 Mha

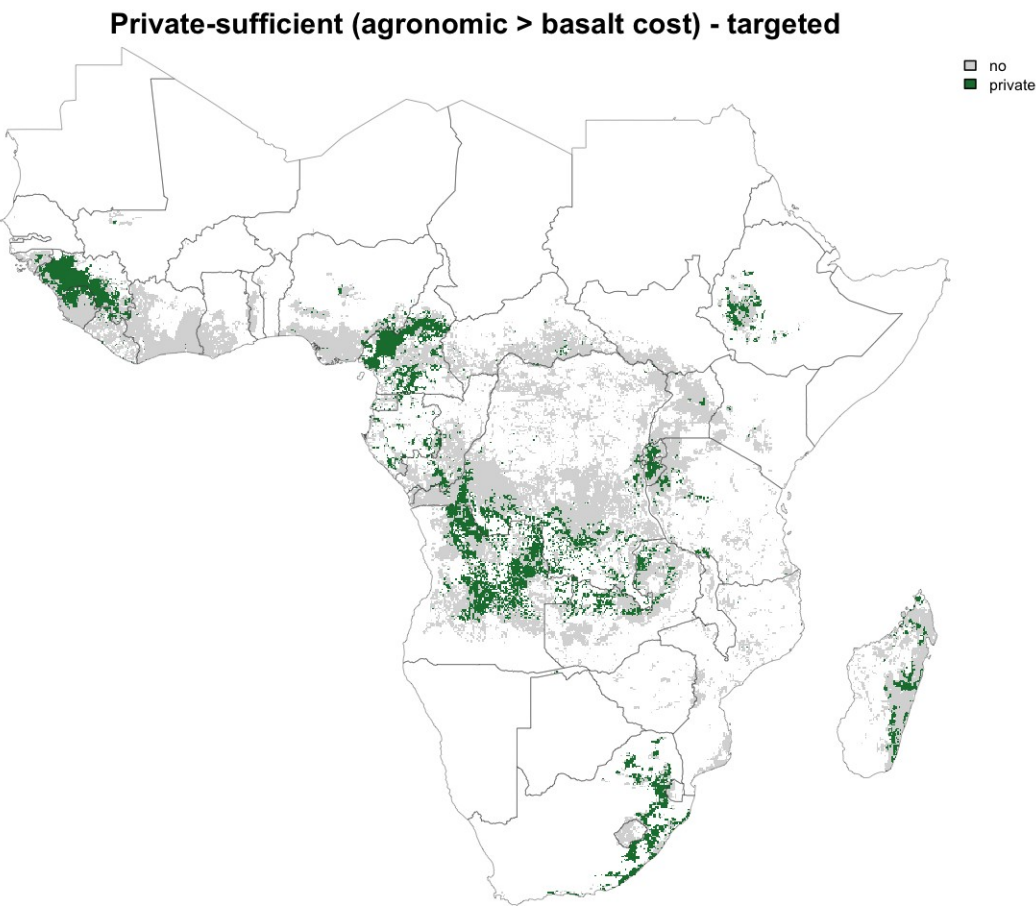
private-sufficient area

\$3.5 B

private return per cycle

The largest single envelope under the targeted dose — and unchanged by the carbon-side model revision, because it does not depend on the CDR surface.

But it is a minority of acid cropland: most viable land needs more than yield to pencil.



Where carbon alone pays (durably)

Land where durable, net-export CDR at \$150/tCO₂ covers the full delivered cost with no yield benefit.

3.95 Mha

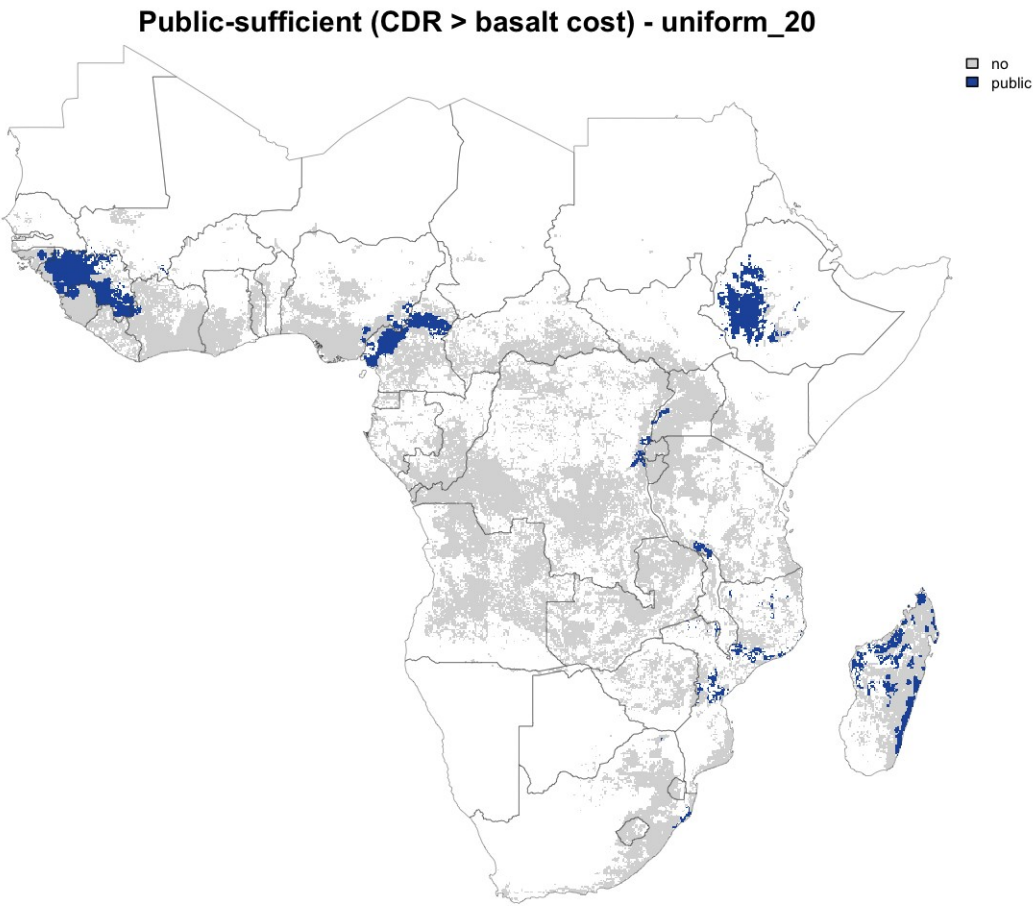
durable public area (20.4 Mt)

12.1%

of treated cropland

On the targeted dose the carbon-only case is 1.95 Mha (11.1% of its treated base) — the lime-requirement dose is CDR-minimal by construction.

Net-export accounting and today's price keep the carbon-only case tightly bound — the reason the combined case matters.



The combined-only band is the policy object

2.7 Mha

combined-only

neither the yield return nor the carbon return alone covers cost — yet together the margin is positive.

4.3

private (Mha)

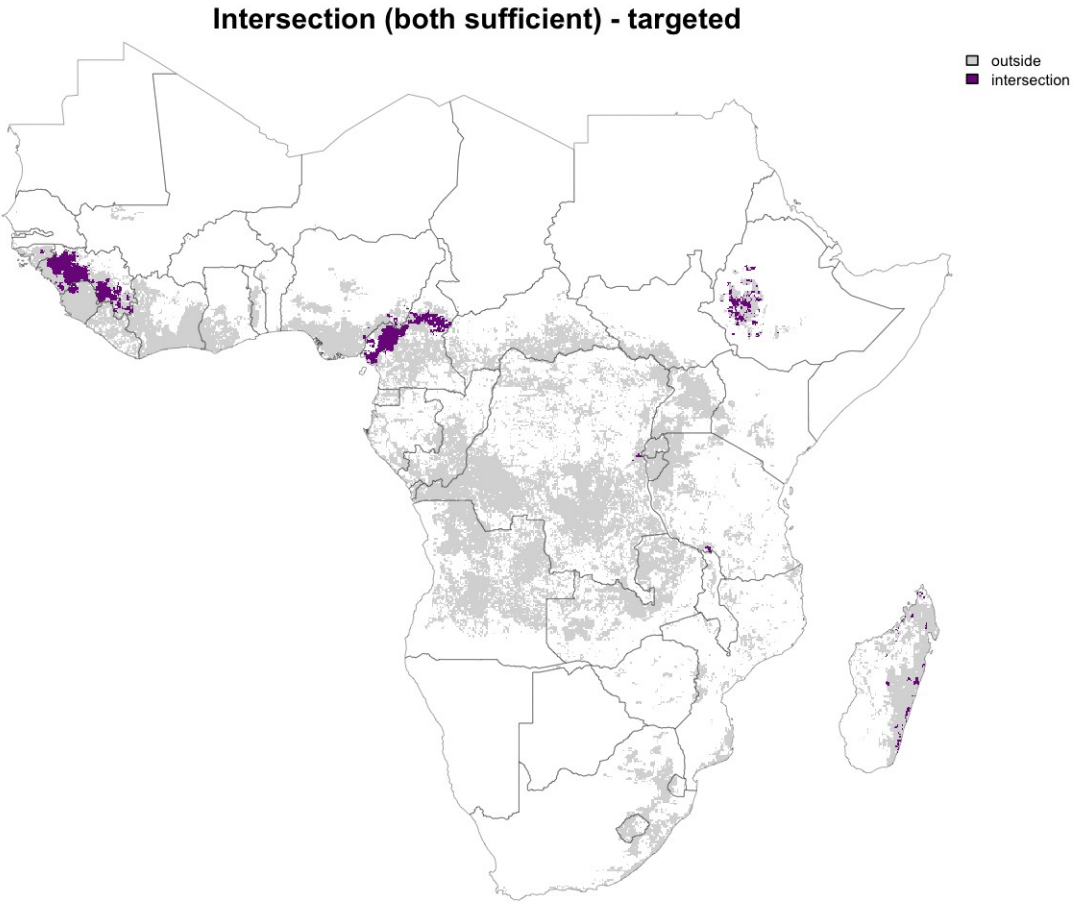
2.0

public (Mha)

7.6

combined (Mha)

The economic point: voluntary carbon finance is not a bonus on an already-profitable practice — it is the marginal revenue that unlocks an otherwise-unprofitable, yield-improving investment.



Marginal abatement cost, not kinetics

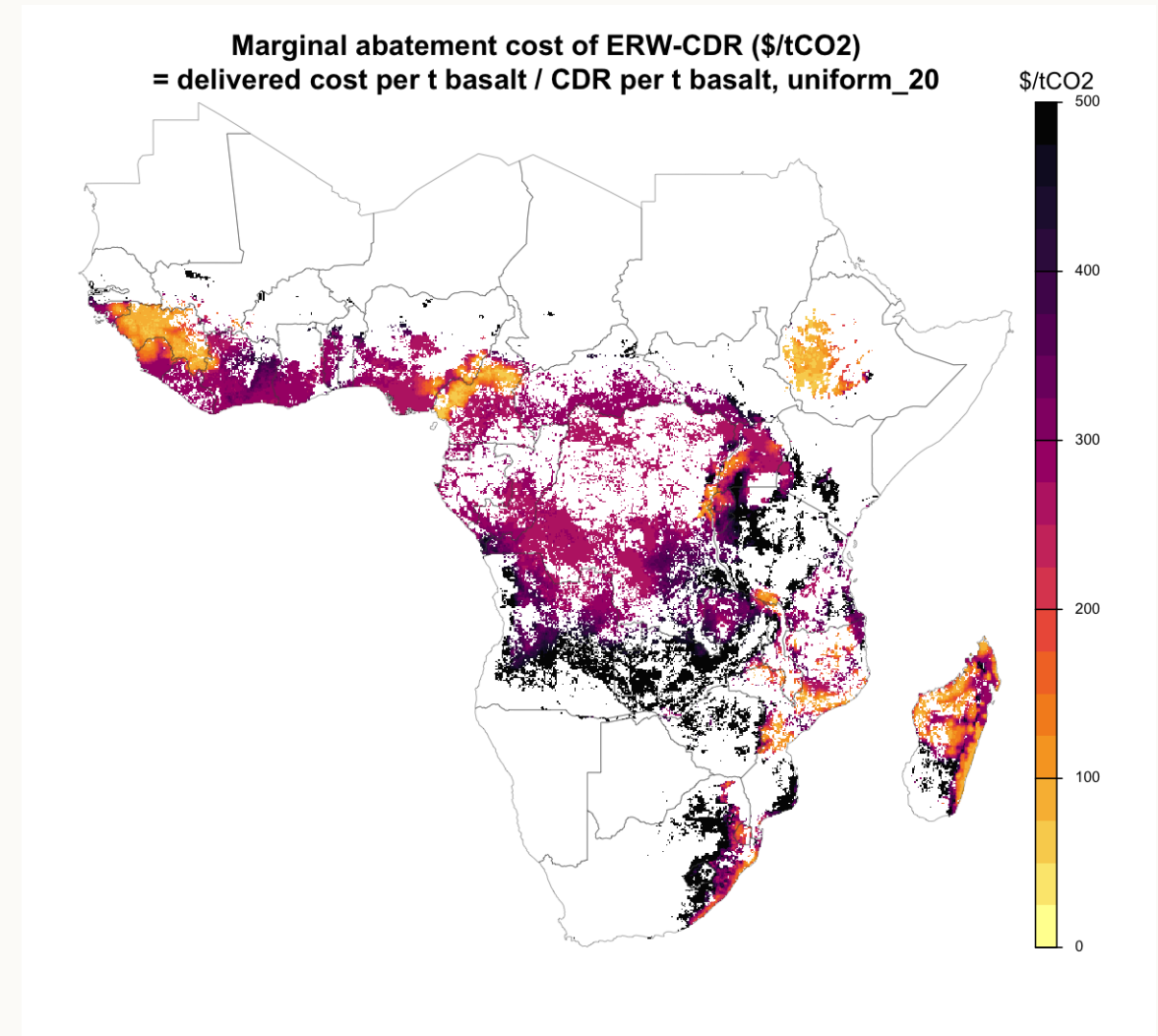
\$276 / tCO₂

area-weighted median MAC — vs a \$130 net credit price
(\$330 on the targeted dose)

MAC = delivered cost per tonne of rock ÷ durable CDR per tonne. The public envelope is exactly the land where MAC < (p - m).

That median sits at the upper edge of the published cost-synthesis IQR (\$137–276; Suhrhoff et al. 2026) — the residual gap is accounting stringency, not SSA logistics.

So the carbon case is limited by **how much CO₂ a tonne of rock verifiably removes** — the largest and least-resolved lever, and the clear field-calibration priority.



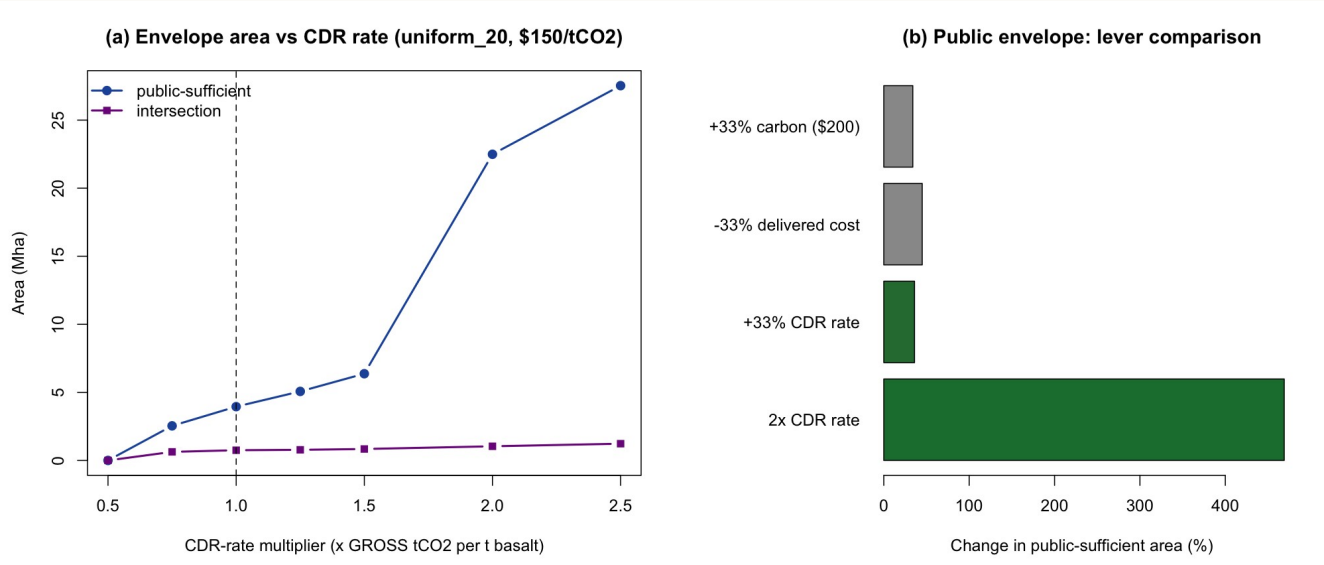
Two levers — not three — shift the public envelope

1 · The price–cost ratio (p - m)/c

Carbon price, MRV and delivered cost enter the public test only through this single ratio: deployable area climbs from 4.3% of treated cropland at \$100 to 86% at \$500. Cost cuts (solar haulage) act on the same lever — and move mostly the private envelope.

2 · Durable CDR per tonne of rock

Scaled where it physically acts — on gross removal, before the acidity sink. Because the sink and life-cycle terms are fixed subtrahends, durable removal is super-linear in it: doubling the rate takes the public envelope 3.95 → 22.5 Mha (~6×). Kinetic saturation is second-order here.

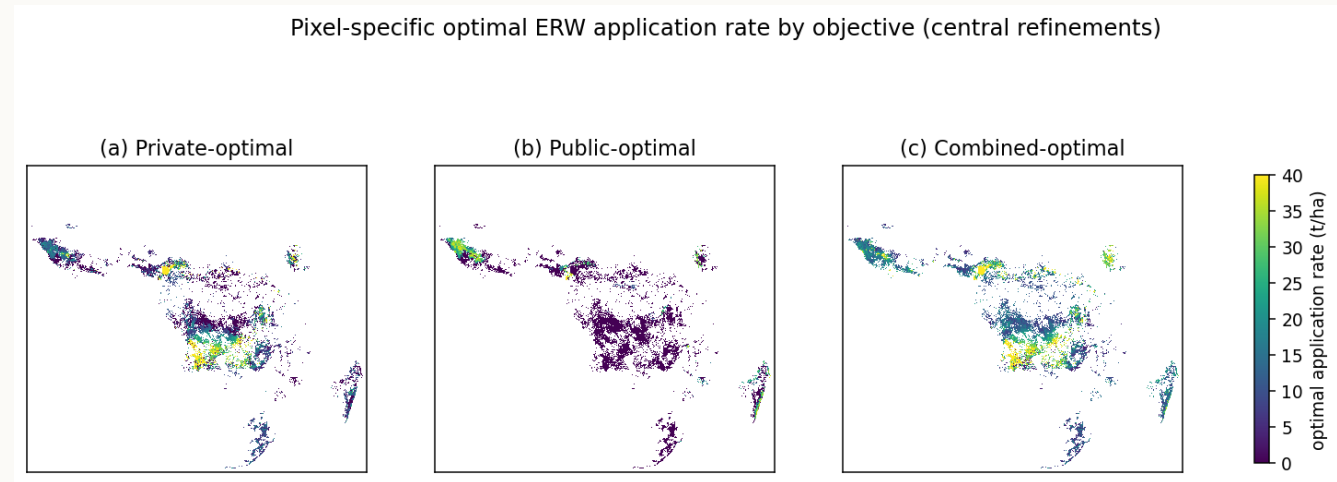


Each objective has an interior optimum

With over-liming and saturation in place, each per-pixel objective is concave — so a pixel-specific **optimal rate** exists for the private, public, and combined margins.

The combined optimum is moderate (typically well below blanket-high uniform doses): enough over-application to earn extra carbon, but not so much that over-liming and saturation erode the return.

Prototype refinement — maps are illustrative (pre-revision basis); promoting them to the headline is on the roadmap.



A defensible priority geography

Intersecting the envelope across the 180-point parameter grid isolates land that stays viable under most assumptions.

0.65 Mha

robust core (≥50% of grid)

0.38 M

farms reached

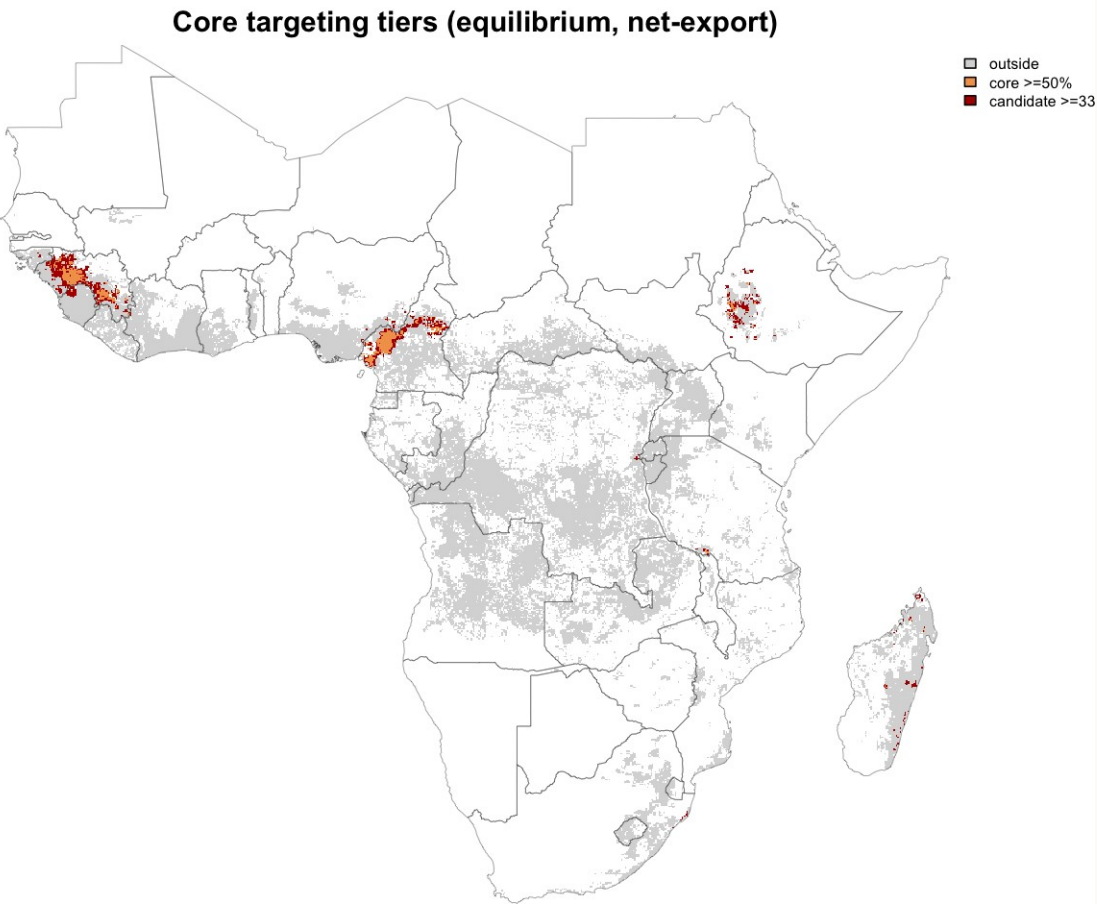
\$0.60 B

crop value at stake

1.22 Mha

candidate tier (≥33%)

Four-fifths of the candidate tier sits in Cameroon (0.61 Mha) and Guinea (0.38) — the geography to deploy first, and the base to expand as the price–cost ratio or the verified CDR rate improves.



ERW is a coupled agronomic-and-carbon proposition

1 · The two returns rarely each suffice alone

But together they make a meaningful area of SSA cropland profitable that neither would — the 2.7 Mha combined-only band.

2 · Carbon finance is the margin, not a bonus

On most viable land, VCM revenue is what unlocks a yield-improving investment. That reframes ERW away from a pure CDR supply curve.

3 · The binding constraint is durable CDR per tonne

Net-export accounting cuts first-application removal ~70%; the area-weighted median MAC is \$276 (uniform-20) / \$330 (targeted). The price–cost ratio and the verified CDR rate are the only two structural levers.

4 · Treating it as coupled changes the answer

Both where one would deploy first (the Cameroon and Guinea highlands core) and what one would invest in to expand it (field-calibrated removal rates and cheaper delivery).

NEXT

Caveats and where we go next

CAVEATS

Priorities

Field calibration of durable removal per tonne is the single highest-value input — it sets MAC and the whole public case.

Not yet modeled

Tenure and upfront finance; compaction; downstream river–ocean losses (~15–20% of exported removal, Renforth & Henderson 2017); country-level credit eligibility (Article 6). Omitted N₂O effects likely run in ERW's favor.

Farm counts

Based on national average holding sizes — order-of-magnitude.

NEXT

Registry-grade accounting

Align the net-export bound with VCM methodology counterfactuals (liming substitution, financial-additionality screens).

Optimal-rate maps

Promote the pixel-specific private/public/combined optima from prototype to headline.

Share & collaborate

The interactive envelope explorer reproduces every slider of this analysis in one self-contained HTML file — no install.